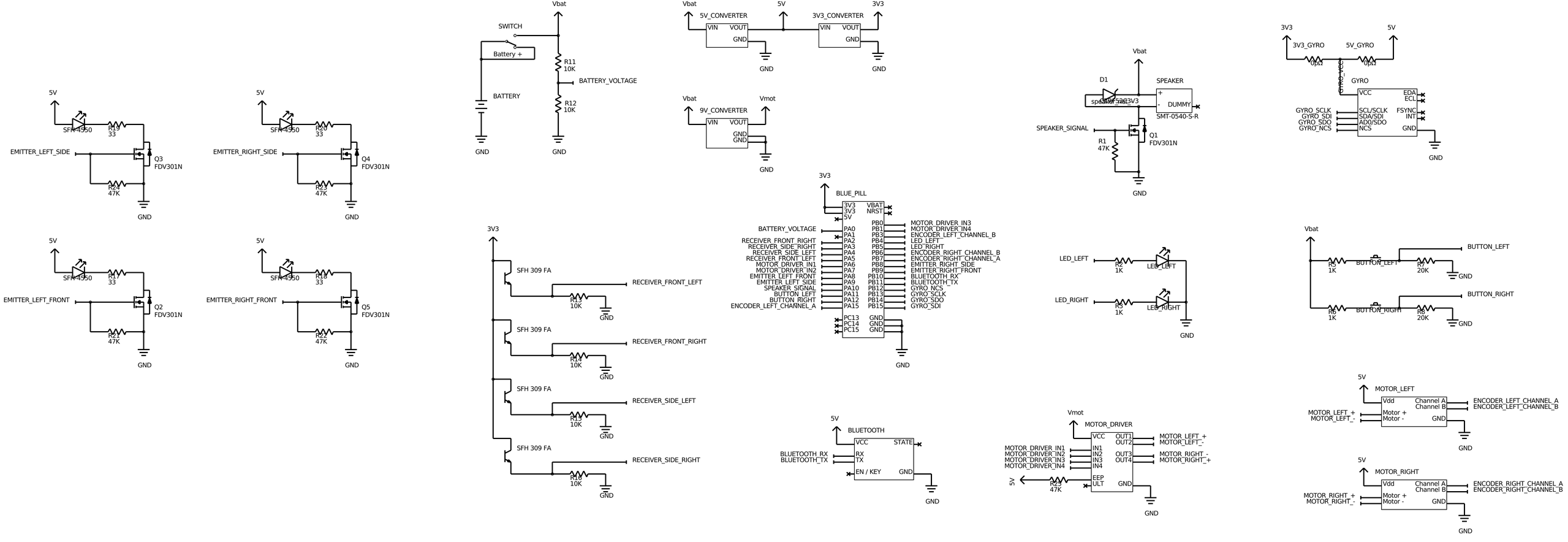
5V
SFH 4550
R19
33
EMITTER_LEFT_SIDE
Q3
FDV301N
R24
47K
GND
5V
SFH 4550
R20
33
EMITTER_RIGHT_SIDE
Q4
FDV301N
R23
47K
GND
5V
SFH 4550
R17
33
EMITTER_LEFT_FRONT
Q2
FDV301N
R21
47K
GND
5V
SFH 4550
R18
33
EMITTER_RIGHT_FRONT
Q5
FDV301N
R22
47K
GND
Vbat
SWITCH
Battery +
R11
10K
BATTERY_VOLTAGE
BATTERY
R12
10K
GND
GND
Vbat
5V_CONVERTER
VIN
VOUT
GND
GND
5V
3V3_CONVERTER
VIN
VOUT
GND
3V3
GND
Vbat
9V_CONVERTER
VIN
VOUT
GND
GND
Vmot
GND
Vbat
D1
SPEAKER
speaker_sig
3V3
+
-
DUMMY
SMT-0540-S-R
SPEAKER_SIGNAL
R1
47K
Q1
FDV301N
GND
3V3
3V3_GYRO
5V_GYRO
0pt1
0pt1
5V
GYRO_VCC
GYRO
VCC
EDA
ECL
GYRO_SCLK
GYRO_SDI
GYRO_SDO
GYRO_NCS
SCL/SCLK
SDA/SDI
AD0/SDO
NCS
FSYNC
INT
GND
GND
3V3
BLUE_PILL
3V3
3V3
5V
VBAT
NRST
BATTERY_VOLTAGE
RECEIVER_FRONT_RIGHT
RECEIVER_SIDE_RIGHT
RECEIVER_SIDE_LEFT
RECEIVER_FRONT_LEFT
MOTOR_DRIVER_IN1
MOTOR_DRIVER_IN2
EMITTER_LEFT_FRONT
EMITTER_LEFT_SIDE
SPEAKER_SIGNAL
BUTTON_LEFT
BUTTON_RIGHT
ENCODER_LEFT_CHANNEL_A
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA15
PC13
PC14
PC15
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
MOTOR_DRIVER_IN3
MOTOR_DRIVER_IN4
ENCODER_LEFT_CHANNEL_B
LED_LEFT
LED_RIGHT
ENCODER_RIGHT_CHANNEL_B
ENCODER_RIGHT_CHANNEL_A
EMITTER_RIGHT_SIDE
EMITTER_RIGHT_FRONT
BLUETOOTH_RX
BLUETOOTH_TX
GYRO_NCS
GYRO_SCLK
GYRO_SDO
GYRO_SDI
GND
GND
3V3
SFH 309 FA
R13
10K
RECEIVER_FRONT_LEFT
GND
SFH 309 FA
R14
10K
RECEIVER_FRONT_RIGHT
GND
SFH 309 FA
R15
10K
RECEIVER_SIDE_LEFT
GND
SFH 309 FA
R16
10K
RECEIVER_SIDE_RIGHT
GND
5V
BLUETOOTH
VCC
STATE
RX
TX
BLUETOOTH_RX
BLUETOOTH_TX
EN / KEY
GND
GND
LED_LEFT
R7
1K
LED_LEFT
GND
LED_RIGHT
R8
1K
LED_RIGHT
GND
Vmot
MOTOR_DRIVER
VCC
OUT1
OUT2
OUT3
OUT4
IN1
IN2
IN3
IN4
EEP
ULT
GND
MOTOR_DRIVER_IN1
MOTOR_DRIVER_IN2
MOTOR_DRIVER_IN3
MOTOR_DRIVER_IN4
MOTOR_LEFT +
MOTOR_LEFT_-
MOTOR_RIGHT -
MOTOR_RIGHT_+
5V
R29
47K
GND
Vbat
R5
1K
BUTTON_LEFT
R7
20K
GND
BUTTON_LEFT
R6
1K
BUTTON_RIGHT
R8
20K
GND
BUTTON_RIGHT
5V
MOTOR_LEFT
Vdd
Channel A
Channel B
Motor +
Motor -
GND
MOTOR_LEFT +
MOTOR_LEFT_-
ENCODER_LEFT_CHANNEL_A
ENCODER_LEFT_CHANNEL_B
GND
5V
MOTOR_RIGHT
Vdd
Channel A
Channel B
Motor +
Motor -
GND
MOTOR_RIGHT_+
MOTOR_RIGHT_-
ENCODER_RIGHT_CHANNEL_A
ENCODER_RIGHT_CHANNEL_B
GND